Customer Churn Risk Report

Report ID: CHR-B93EE35974 Generated: 18 October 2026, 21:05

Prediction

Likely to stay - churn probability **5.9%**

Risk level	Low
Decision threshold	55.0% (above this = likely to churn)
Retention probability	94.1%

The model estimates a low risk of leaving. No urgent action is indicated.

Customer details

Contract type	Two year
Payment method	Bank transfer (automatic)
Internet service	DSL
Online security	Yes
Tech support	Yes
Tenure	58 months
Monthly charges	$45.00
Total charges	$2,610.00

Key factors

Input	Value	Effect on estimate
Contract type	Two year	Pushes the estimate down
Online security	Yes	Pushes the estimate down
Tech support	Yes	Pushes the estimate down
Tenure	58 months	Pushes the estimate up
Payment method	Bank transfer (automatic)	Pushes the estimate down

Factors describe patterns the model associates with churn, not proven causes.

Recommended retention actions

• Schedule an early-life check-in; newer customers leave more often in this data.

Model information

Model	Gradient Boosting
Evaluation	1409 held-out customers (stratified 20% test split, seed 42)
Accuracy / precision / recall	76.2% / 53.7% / 75.4%
F1-score	62.7%
ROC-AUC	0.838
PR-AUC (baseline = churn rate)	0.657 (baseline 0.265)
Brier score (lower is better)	0.164 (baseline 0.195)
Confusion matrix (TN/FP/FN/TP)	792 / 243 / 92 / 282

Data	Telco-Customer-Churn.csv (7,043 customers)

How to read the probability

• The predicted outcome compares the estimated probability with the decision threshold.

• Risk bands: Low below 40%, Moderate from 40% to below 70%, High from 70%.

• Key factors list the inputs that moved this customer's estimate the most.

Limitations

• The model was trained on one public telecom dataset (IBM Telco Customer Churn). Results may not transfer to other businesses, regions or time periods.

• The model does not see support history, usage, contract dates or competitor offers, which often matter for churn.

• Key factors show patterns the model associates with churn. They are not proven causes.

• Probabilities are estimates. Compare customers with each other rather than reading any single score as an exact frequency.

This report is generated by a statistical model trained on a public telecom dataset. The churn probability is an estimate of how similar customers behaved in that data. It is not a guarantee of what this customer will do, and it does not prove why they would leave. Use it together with your own knowledge of the customer.